

# Classifying non-stationary time-varying signals on graphs through their deviation from a stationary behavior: preliminary results on EEG signals

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## I. INTRODUCTION

Brain Computer Interfaces systems in general aspect are systems that transform brain signals into control signals. In this study, we focus on electroencephalography (EEG) signals for brain signal acquisition, which are known to be high-dimensional, non-stationary and noisy.

We consider here the classification problem of finding the Motor Imagery (MI) class — *e.g.*, mental visualization of a move of a hand or a foot — associated with a given EEG signal. Classical approaches for tackling this problem include Common Spatial Pattern [1], Tensor methods [2], or Divergence methods [3]. Because these solutions do not consider the structure of EEG signals, newer approaches based on graph signal processing have been developed [4]. While many of these techniques assume stationarity and/or smoothness of graph signals — which does not match the signal model considered here —, some recent approaches have started to consider more complex non-stationary signals on graphs [5], [6]. Still, none of these methods assume signals to be both non-stationary on graphs, and time-varying.

In this work, we propose to discuss a possible *regularization* of non-stationarity, by studying the deviation of an EEG signal from its stationary behavior on a graph that we model by the covariance matrix of the electrodes. The postulate of this article is that signals from different classes should deviate from their stationary behavior in distinct manners. We show here our first observations on this methodology.

## II. METHODOLOGY

We consider here the *motor imagery, uncued classifier application* dataset from the Berlin Brain-Computer Interface group<sup>1</sup>, consisting of EEG MI signals [7] sampled at 1,000Hz for 3.5s. Let  $\mathcal{D} = \{\mathbf{D}_1, \mathbf{D}_2, \dots, \mathbf{D}_E\}$  be our training set of  $E$  time-varying EEG signals, each element of the set being a  $N \times T$  matrix of measurements of  $N$ -dimensional signals for  $T$  consecutive units of time. Let  $\mathcal{C} = \{c_1, c_2, \dots, c_E\}$  be the MI classes associated with the elements in  $\mathcal{D}$ . When considering a two-class problem, we have  $\{c_i\}_i \in \{\text{hand}, \text{foot}\}$ .

Let us consider an EEG signal  $\mathbf{D}_i \in \mathcal{D}$ . We know from [8] that the covariance matrix of stationary signals on graphs has

the same eigenvectors as a matrix representing the graph on which they evolve. We choose to analyze the deviation of  $\mathbf{D}_i$  to its stationary behavior as follows:

- 1) We extract submatrices of  $\mathbf{D}_i$  using a time-varying sliding window, so that we obtain a series  $(\mathbf{D}_i^{1:W}, \mathbf{D}_i^{S:W+S}, \dots, \mathbf{D}_i^{T-W:T})$ , where  $W$  is the window size and  $S$  is the stride. Each of these elements is a matrix of dimensions  $N \times W$ .
- 2) For each of the elements  $\mathbf{D}_i^j$  of this series, we compute its  $N \times N$  sample covariance matrix  $\tilde{\Sigma}_i^j$ .
- 3) We compute the average sample covariance matrix  $\tilde{\Sigma}_i$  of  $(\tilde{\Sigma}_i^j)_j$  using some metric  $d$ :

$$\tilde{\Sigma}_i = \arg \min_{\Sigma \in \mathcal{M}_{\text{spd}}} \sum_j d(\Sigma, \tilde{\Sigma}_i^j), \quad (1)$$

where  $\mathcal{M}_{\text{spd}}$  is the manifold of symmetric positive definite matrices.

- 4) We compute the projections  $(\tilde{\mathbf{A}}_i^j)_j$  of the eigenvalues of the matrices  $(\tilde{\Sigma}_i^j)_j$  in the eigenbasis of  $\tilde{\Sigma}_i$  as follows:

$$\forall j : \tilde{\mathbf{A}}_i^j = \tilde{\chi}_i^\top \tilde{\Sigma}_i^j \tilde{\chi}_i, \quad (2)$$

where  $\tilde{\chi}_i$  are the eigenvectors of  $\tilde{\Sigma}_i$  (see [8] for additional details).

- 5) Because the eigenvectors of  $(\tilde{\Sigma}_i^j)_j$  are not aligned with  $\tilde{\chi}_i$  due to non-stationarity of signals and a limited number of samples,  $(\tilde{\mathbf{A}}_i^j)_j$  is not in general a diagonal matrix. Therefore, we approximate it as follows:

$$\forall j : \tilde{\Lambda}_i^j = \arg \min_{\Lambda \in \mathcal{M}_{\text{diag}}} \rho(\mathbf{I} - \Lambda^{-1} \tilde{\mathbf{A}}_i^j), \quad (3)$$

where  $\mathbf{I}$  is the identity matrix, and  $\mathcal{M}_{\text{diag}}$  and  $\rho(\cdot)$  are the space of diagonal matrices and the spectral radius respectively.

Since we consider sliding windows, we expect two consecutive submatrices of  $\mathbf{D}_i$  to have slightly different eigenvalues when seen from the same eigenbasis  $\tilde{\chi}_i$ . Therefore, the series  $(\tilde{\Lambda}_i^{1:W}, \tilde{\Lambda}_i^{S:W+S}, \dots, \tilde{\Lambda}_i^{T-W:T})$  should form some trace in an  $N$ -dimensional space that we expect to be characteristic of the class  $c_i$ .

By doing so, we are performing some sort of regularization to study deviation of a time signal on a graph from its stationary behavior on a graph.

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<sup>1</sup> [http://www.bbc.de/competition/iv/desc\\_1.html](http://www.bbc.de/competition/iv/desc_1.html)

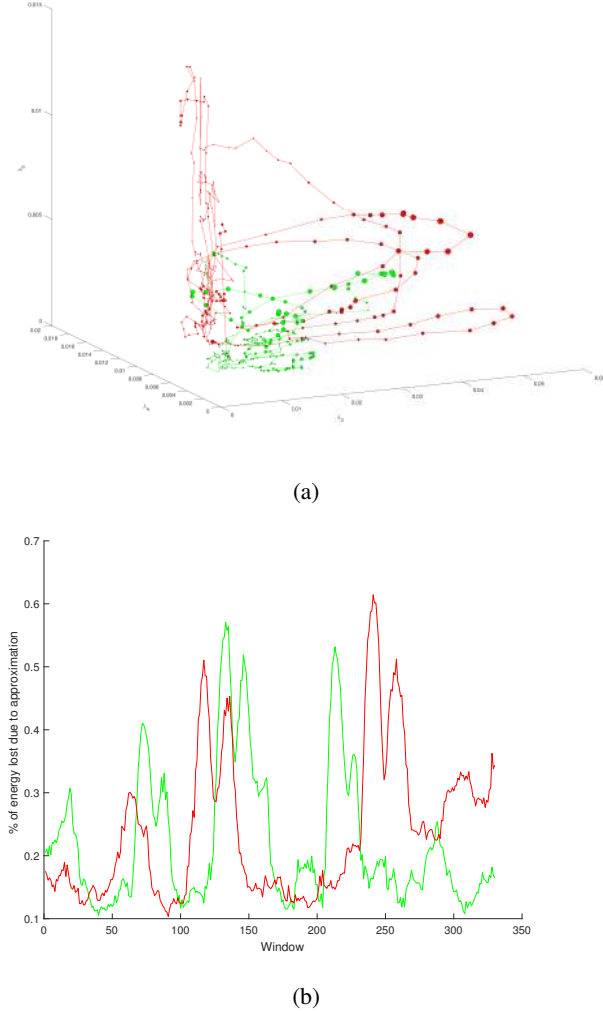


Figure 1: Examples of two traces associated with one EEG signal of each of the classes, *foot* in green and *hand* in red **(a)**. Each point represents the approximate eigenvalues of the sample covariance matrix of one time window  $\tilde{\Sigma}_i^j$  on the EEG signal. For visualization purpose, we restrict to three eigenvalues. The size of the point represents the approximation error  $\frac{\|\tilde{\mathbf{A}}_i^j - \text{diag}(\tilde{\mathbf{A}}_i^j)\|_F}{\|\tilde{\mathbf{A}}_i^j\|_F}$  for the eigenvalues of  $\tilde{\Sigma}_i^j$  **(b)**, which can be interpreted as the deviation of the corresponding window to the stationary behavior of the whole EEG signal on the graph. Points associated with consecutive windows are linked with an edge, defining a trajectory for the estimated eigenvalues, that we expect to carry some information on the class of the EEG signal.

### III. FIRST OBSERVATIONS

For each element  $\mathbf{D}_i \in \mathcal{D}$ , we compute its trace  $(\tilde{\mathbf{A}}_i^{1:W}, \tilde{\mathbf{A}}_i^{S:W+S}, \dots, \tilde{\mathbf{A}}_i^{T-W:T})$ , for  $W = 200$  and  $S = 10$ . Results for two EEG signals are given in Figure 1. Interestingly, the peaks in the approximation error curve appear to match some patterns that appear in the EEG signals. Figure 2 gives the EEG signal associated with the green curve.

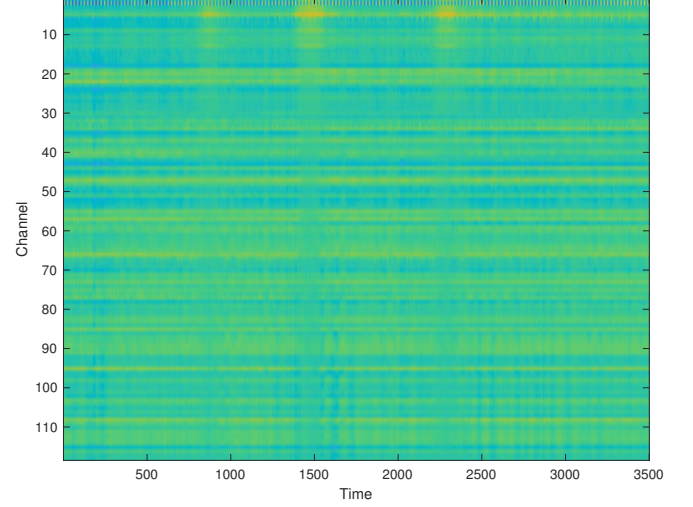


Figure 2: EEG signal associated with the green curve in Figure 1. Peaks in this curve appear to match particular activities in the graph of electrodes around times  $t \in \{900, 1500, 2300\}$ .

It appears that the measure of deviation to a stationary behavior on a graph may help identify moments in an EEG signal that may be characteristic of a particular activity. The direction of our work is to exploit this observation to analyze selected portions of the traces in Figure 1a to find particular features that may be discriminant of the class associated with the EEG signal. We expect to find some features on these portions of the traces — such as Euclidean length, distance to barycenter, etc. — to be significant for classes separation.

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